

AADITYA NEUPANE

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EDUCATION

BS in Artificial Intelligence

Oct 2023 – Present

Deggendorf Institute of Technology (DIT), Deggendorf, Germany

Relevant Coursework: Web Development, Databases, Computer Vision, Machine Learning, Speech Recognition, NLP, RNN, Transformers, LLMs, GAN, Cloud computing, RAG, Multi Agent Systems, AI workflow, Robotics

WORK EXPERIENCE

Summer Researcher | DIT | Deggendorf

Aug 2026 – Present

- Collaborated with domain experts (energy engineers) to translate complex physical building data into actionable AI solutions.
- Built and trained machine learning models to predict energy consumption and optimize efficiency in smart building systems.
- Processed large-scale, real-world time-series sensor data, creating data pipelines for training.

Data Science Intern | Daimler Truck AG | Stuttgart

Oct 2025 – Mar 2026

- Developed an ML and Statistical based data quality monitoring system to detect anomalies in vehicle telematics data.
- Designed data quality metrics and KPIs for ETL pipelines processing 100K+ Mercedes-Benz Truck telematics records.
- Analyzed large scale datasets and built dashboards in Databricks and Power BI to debug and monitor data-quality issues.
- Maintained internal libraries, automated tests, and production data pipelines.

Student Research Assistant | DIT | Deggendorf

Apr 2024 – Mar 2026

- Participated in AI and ML research projects.
- Experimented with diverse Transformation and Feature Extraction techniques on data collected from robots / edge devices
- Engineered scalable MLOps pipeline for large scale Time Series Data.
- Developed CNN based Computer vision models for Industrial Sorting Line Prototype (FischerTechnik Industrie 4.0 and Sorting Line)
- Developed GAN based synthetic data generation pipeline to create realistic electronic signal datasets
- Created data analysis reports and model performance reports for research and stakeholder review.

Student Teaching Assistant | DIT | Deggendorf

Oct 2024 – Sep 2025

- Coded in LaTeX to create documents, presentations and converted the existing documents to a LaTeX format.
- Collaborated with the teams of Professors and Students to enhance teaching materials.

TECHNICAL SKILLS

- Python, OOP, OpenCV, TensorFlow, PyTorch, Matplotlib, Pandas, PySpark, NumPy, Scikit-learn, NLTK, getML, Dask
- Databricks, Jupyter Notebook, Google Colab, Kaggle
- HTML, CSS, JavaScript, Express, React, Three.js
- CI/CD, GitHub, Gitlab, Docker, AWS, Azure DevOps, Jira
- C/C++, Micro-Python with Arduino, Esp32, Raspberry Pi
- SQL, PostgreSQL, FastAPI, MCP
- R and R studio, PowerBI, Databricks Dashboard
- Linux (Ubuntu), PowerShell, Bash
- Hugging face, Ollama, LangChain, LangGraph, N8n
- LaTeX, BibTeX, Citavi, Overleaf
- Microsoft Azure AI Studio, Google Ai Studio

LANGUAGES

- English: C1 (Full Professional Proficiency)
- Nepali: Native speaker
- German: B1 (Intermediate proficiency)

PROJECTS

3D Portfolio Website | React, Three.js, Vite, Express, Next.js, GSAP

- Built a personal portfolio website combining a [3D interactive portfolio](#) with a [VS Code style overlay interface](#).
- Developed the 3D frontend using React, Three.js, Vite, and GSAP, with Express serving the production build.
- Created a separate Next.js based VS Code portfolio interface to showcase projects, skills, and profile content.

Multi-Agent Software Engineering Team | Python, LangGraph, FastAPI, Pydantic AI, MCP, React, TypeScript, Vite

- Currently developing an agentic AI harness that has specialized software engineering agents to transform product ideas into requirements, architecture, code, tests, documentation, and review reports.
- Designing multi agentic workflow with LangGraph for software development workflows, including planning, backend generation, frontend generation, testing, documentation, and “human in the loop” review.
- Building a FastAPI backend and React & TypeScript frontend for agent workflows, and to manage task execution

Local Industrial RAG | Python, LangChain, ChromaDB, FAISS, Ollama, Gemma 4B, Sentence Transformers

- Built a fully offline industrial RAG & CAG system to query technical documentation, manuals, and reference files using a locally hosted Gemma 4B model.
- Developed a document ingestion pipeline to load multi format files, split them into chunks, generate embeddings, and store searchable knowledge in ChromaDB.
- Implemented semantic caching with FAISS to detect similar queries, reuse previous answers, reduce repeated LLM calls, and improve response efficiency.

Titanic Survival Prediction App | React, Docker, Python, Vite, FastAPI, Nginx, Jira, Agile, PostgreSQL, GitLab CI/CD, E2E

- Machine Learning based Full Stack Web App that predicts user’s chance of survival in Titanic
- Developed ML prediction and training APIs and Containerized frontend, backend, model service, database in Docker.
- Led a 7-member team, coordinated development tasks, and managed project workflow using Jira.

Autonomous Vehicles System | PyTorch, OpenCV, TorchVision, TensorRT, JetRacer, Jupyter Notebook

- Built an object detection based system for traffic management in autonomous vehicles.
- Trained a ResNet18-based model using PyTorch to predict steering coordinates from road images.
- Integrated camera input, TensorRT inference, OpenCV logic, and JetRacer control for real-time autonomous driving.

Stroke Predictor and Chatbot Web App | Python, Streamlit, Rasa, XGBoost, scikit-learn, Pandas, Matplotlib, Seaborn

- Built an NLU-based healthcare ML application for stroke risk prediction and a personalized recommendation system.
- Integrated an XGBoost prediction model with a Streamlit interface and Rasa chatbot, including data preprocessing, analysis, and visualization.

AI Based Quality Assurance for FischerTechnik Sorting Line | TensorFlow, TFLite, OpenCV, scikit-learn, XGBoost, tsfresh, getML

- Worked as a Student Research Assistant (SHK) on a university’s AI based quality assurance project, contributing to [computer vision model optimization](#) and [Sensor Data based ML analysis](#) for a FischerTechnik sorting line.
- Improved Computer Vision based quality inspection for the FischerTechnik sorting line by training and testing image classification and object detection models using TensorFlow Lite, MobileNetV2, EfficientDet-Lite0, and OpenCV.

Anti Distraction Extension | JavaScript, Html, CSS

- Chrome Extension that Replaces Distractive videos on YouTube with a Motivational Phrase "Study hard".

AI Presentation Generator | N8n, Gemini Flash, Streamlit, Unsplash

- Built a N8N workflow & Streamlit app that automatically generates presentations from a topic, which won a hackathon.

CERTIFICATIONS / COURSES

LangChain Academy: Deep Agents

Jun 2026

LangChain Academy: Deep Research with LangGraph

Jun 2026

COMMUNITY INVOLVEMENT

Gesellschaft für Informatik, *Member*

Jan 2024 – Present

University Clubs: Chess, Swimming

Oct 2023 - Present